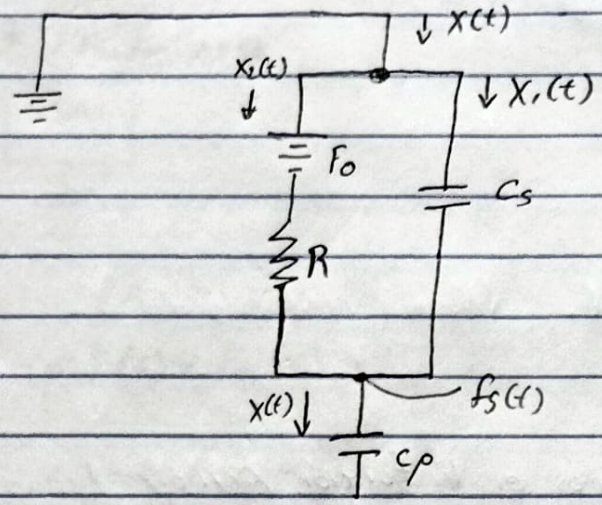
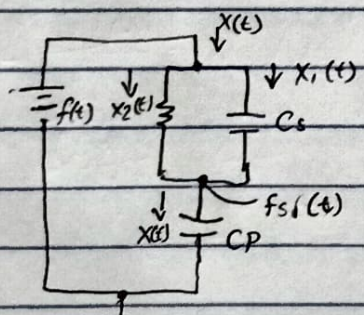
Practica #3 (5.5) Modelado y control del sistema musculoesquelético

Transformación de sistema mecanico a electrico.

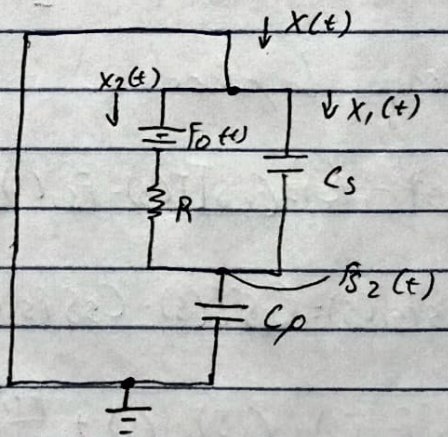


$F_s(t) =$

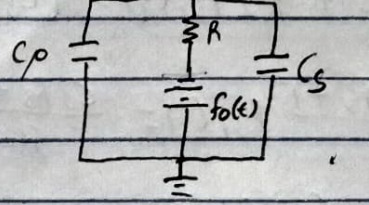
* Superposición ($F_{s1}(t)$)

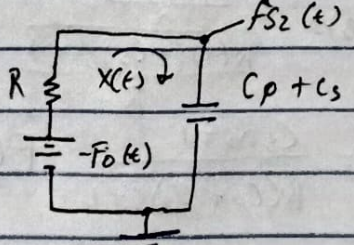


* Superposicion ($F_{s2}(t)$)



Reduccion del circuito.



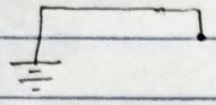


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* Analisis matematico.

Debido a que el sistema contiene dos fuentes de voltaje, se debe aplicar el principio de superposición.

- Circuito #1 / apagando $F_0(t)$



$X(t) = X_1(t) + X_2(t)$

$X(t) = C_p \frac{dF_{s1}(t)}{dt}$

$X_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$

$X_2(t) = \frac{F(t) - F_{s1}(t)}{R}$

* Sustituimos en la función principal

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

* Aplicamos Laplace:

$$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) + C_s s F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

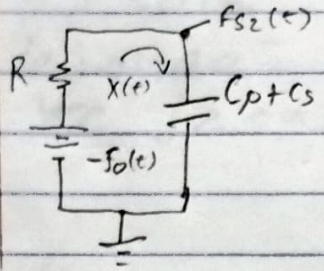
$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s)$$

$$F_{s1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

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* Circuito #2: Apagando $F(t)$

Recordando que $F_o(t) = \alpha F(t)$



* Análisis:

$$-\alpha F(t) = R x(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

* Laplace:

$$-\alpha F(s) = R x(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{x(s)}{(C_p + C_s)s}$$

* Función de Transferencia Segundo Circuito: $\frac{F_{s2}(t)}{-\alpha F(s)}$

$$F(s) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_p + C_s)s}\right) x(s) = \frac{1}{\alpha} \cdot \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}\, x(s)$$

$$\frac{F_{s2}(s)}{F(t)} = \frac{\frac{x(s)}{(C_p + C_s)s}}{-\frac{1}{\alpha}\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}\, x(s)} = -\frac{\alpha F(t)}{R(C_p + C_s)s + 1}$$

* Función de transferencia:

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1} F(s) \quad \therefore \quad \frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}$$

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* Error en estado estacionario:

$$e(s) = \lim_{s \to \emptyset} s F(s) \left(1 - \frac{F_s(s)}{F(s)}\right) = s \frac{1}{s} \left(1 - \frac{RCs\,s + 1 - \alpha}{(RCp + RCs)s + 1}\right)$$

$$e(s) = 1 - \frac{1-\alpha}{1} = 1 - 1 + \alpha = \alpha$$

* Estabilidad en lazo abierto:

$$(RCp + RCs)s + 1 = \emptyset$$

$$\lambda = -\frac{1}{RCp + RCs} = \text{Sistema estable.} \quad (\text{Numero real negativo})$$

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